

Game Theory

ENEL445 Applied Engineering Optimisation

2025

Game Theory

- Game theory is the mathematical analysis of decision making.
- The interaction between two or more players is framed in terms of a **game** with a particular set of rules.
- Of interest may be the strategies that give optimal outcomes for each of the players or, conversely, the resulting outcomes when certain strategies are played.
- Game theory came to prominence in the latter half of the twentieth century. Has led to several Nobel Prizes in economics, as well as significant advances in biology, computer science, and political science.

Game Theory

- Examples of games:
 - Driving game - Drivers manoeuvring in heavy traffic
 - Auctioning game - Shoppers bidding on TradeMe
 - Bargaining game - Unions negotiating
 - Political game - Candidates in an election
 - Economic game - Supermarkets deciding prices
- A game is being played whenever humans interact!

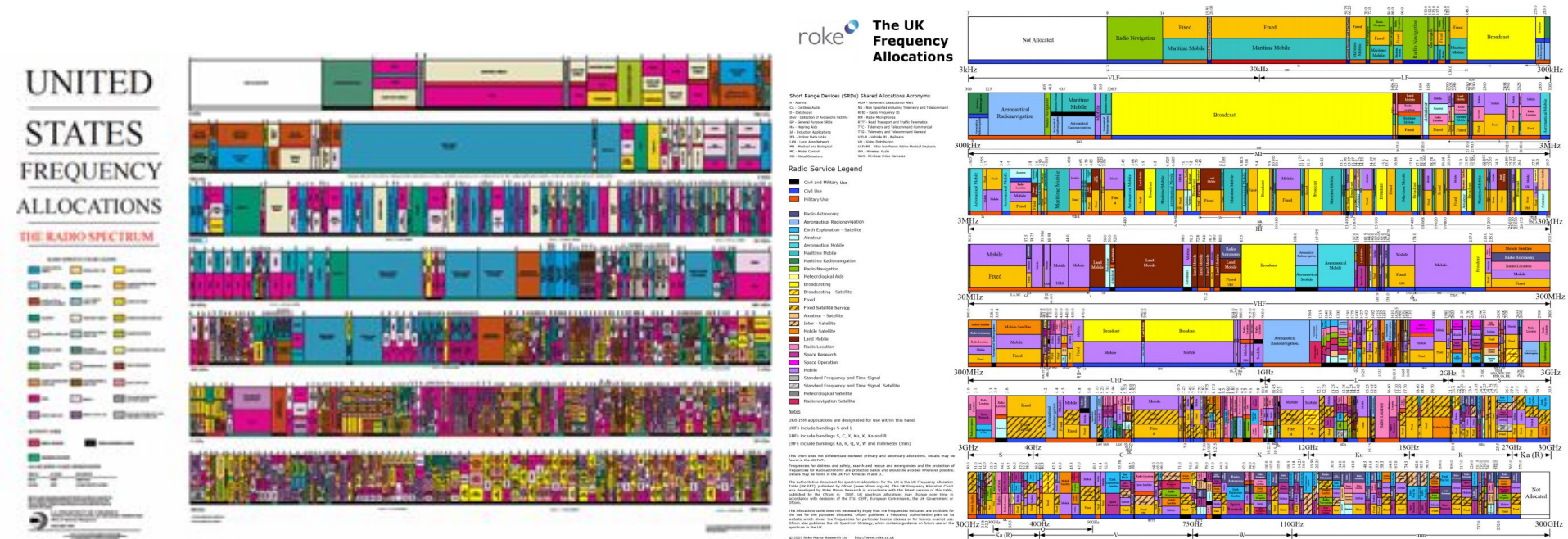
“Antony and Cleopatra played the courting game on a grand scale. Bill Gates made himself immensely rich by playing the computer software game. Hitler and Stalin played a game that killed off a substantial fraction of the world’s population. Khrushchev and Kennedy played a game during the Cuban missile crisis that might have wiped us out altogether.”

Game Theory

- With such a wide field of application, game theory would be a universal panacea if it could always predict how people will play the many games of which social life largely consists.
- Game theory isn't able to solve all of the world's problems because it only works when people play games **rationally**.
- So Game theory can't predict the behaviour of love-sick teenagers or crazy political persons.
- However people don't always behave irrationally. Most of us at least try to spend our money sensibly - and we don't do too badly most of the time or economic theory wouldn't work at all.

Game Theory

- Game theory was applied to design the rules that were used for the US and British governments to auction off the right to various radio frequencies for use with cell phones.
- The result was that the American taxpayer made a profit of \$20 billion - more than twice the prediction. Even more was made in a later British telecom auction of \$35 billion in just one auction.



Example - Prisoner's Dilemma

- The police have captured two criminals and are interrogating them in separate rooms, so they can't communicate with each other. The police offer each the following deal:
 - If A snitches on B, A goes free and B spends 20 years in jail. (A defects, B cooperates)
 - If B snitches on A, B goes free and A spends 20 years in jail. (A cooperates, B defects)
 - If neither of them snitch on each other, then they both spend one year in jail. (Mutual cooperation)
 - If they both snitch on each other, then they both spend 5 years in jail. (Mutual defection)

Payoff
Matrix

		Prisoner B	
		Confess	Not Confess
Prisoner A	Confess	5, 5	0, 20
	Not Confess	20, 0	1, 1

Example - Prisoner's Dilemma

- Assume that there is no “honour among thieves” and each prisoner only cares about minimising their own jail time.
- Let's assume the role of A. We're looking to minimise our prison time. Since we have no way of knowing whether our partner in crime has confessed, let's first assume that they have not. If A doesn't confess either, both will go to prison for 1 year. Not bad. But, if A confesses, he will go free, while his partner rots away in jail. From the above discussion, it is obvious that if B does not confess, A is definitely better off confessing.

		Prisoner B	
		Confess	Not Confess
Prisoner A	Confess	5, 5	0, 20
	Not Confess	20, 0	1, 1

Example - Prisoner's Dilemma

- Now let's look at the other possibility. Say B confesses. If A does not confess, he will go to jail for 20 years. But if he does confess, he will get only 5 years in prison. It is clearly better to confess in this case as well.
- Is each prisoner better off confessing? It may seem so from the above discussion, but if we look at the payoff matrix, it is clear that the best payoff for both prisoners is when neither confesses! But game theory advocates that both confess.

		Prisoner B	
		Confess	Not Confess
Prisoner A	Confess	5, 5	0, 20
	Not Confess	20, 0	1, 1

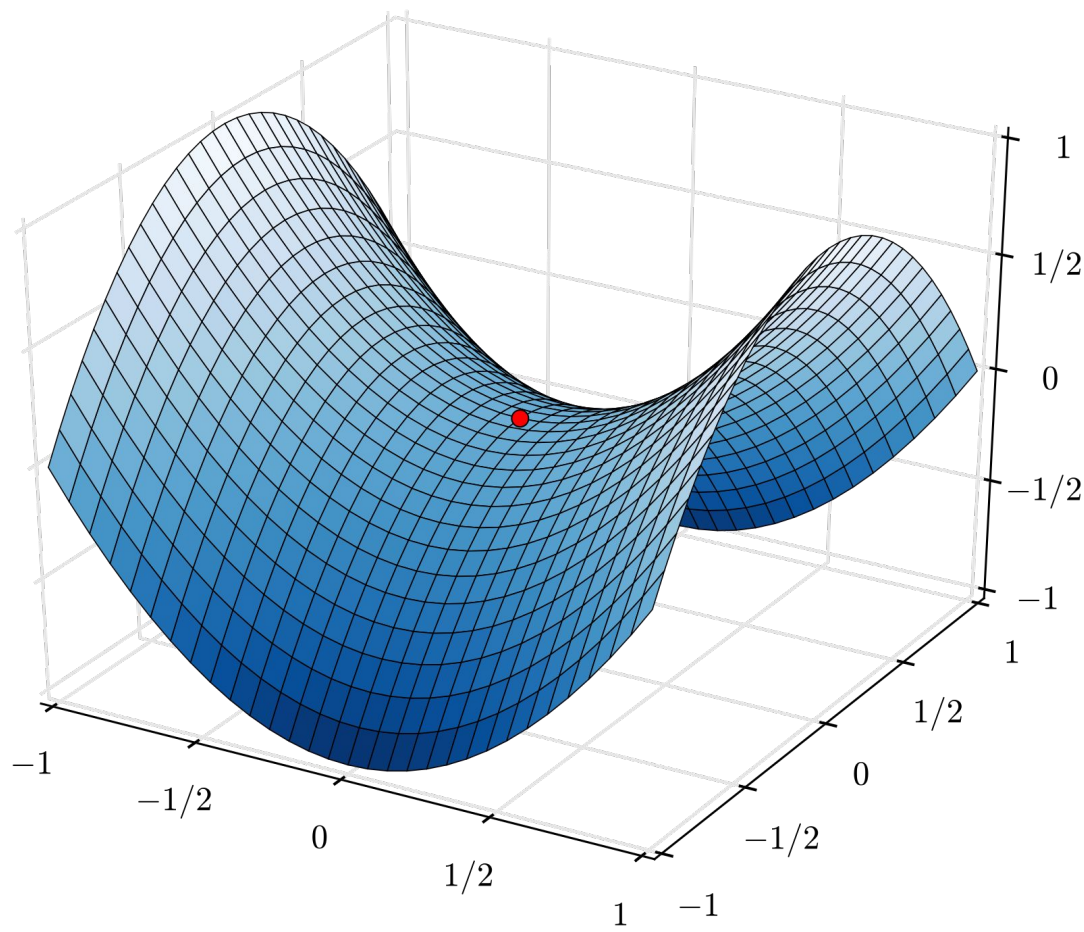
Example - Prisoner's Dilemma

- Can be generalised to any situation when two players are in a non-cooperative situation where the best all-around situation is for both to cooperate, but the worst individual outcome is to be cooperating while the other player defects.
- On the one hand, it is tempting to defect, or confess. Since you have no way of influencing the other player's decision, no matter what they do, you're better off confessing. But on the other hand, you're both in the same boat. Both of you should be sensible enough to realise that cheating undermines the common good.
- There is no single "right" solution to the Prisoner's Dilemma (that's why it's a dilemma). Its implications carry into psychology, economics, and many other fields.

Minimax

- Formally, the minimax theorem states for a concave-convex function f :

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y)$$



Maxi mini things

- Maximax
- Maximin
- Minimax
- Minimin(?)

Maximax

- Choose the **best of the best** possible outcomes.
- Example Game: Two players simultaneously put either a blue or a red card on the table.
 - If player 1 puts a **red card** on the table, whichever card player 2 puts down, no one wins anything.
 - If player 1 puts a **blue card** on the table and player 2 puts a **red card**, then player 2 wins \$1000 from player 1.
 - If player 1 puts a **blue card** on the table and player 2 puts a **blue card** down, then player 1 wins \$1000 from player 2.

		Player 2	
		Blue	Red
Player 1	Blue	1,000	-1,000
	Red	0	0

Maximax

- Applying the maximax principle: Player 1 will always play the blue card, since it has the maximum possible payoff of 1000. But as can be clearly seen from the table, assuming player 2 is rational, they will never play the blue card, since the red card gives him the dominant strategy. In such a case, if player 1 plays by the maximax rule, player 1 will always lose.
- The maximax principle is inherently irrational, as it does not take into account the other player's possible choices. Maximax is often adopted by naive decision-makers.

		Player 2	
		Blue	Red
Player 1	Blue	1,000	-1,000
	Red	0	0

Maximin

- Choose the **best of the worst** possible outcomes.
- Maximin is a **one-person game** strategy.
- Example Game: A player has only two choices to make - to gamble or not to gamble:

		Chance	
		Win	Lose
Player	Gamble	10000	-1
	Not gamble	0	0

- Applying the maximin principle: The player should never gamble, because he faces a risk of losing \$1.

Minimax - One-person

- Example Game: A company is trying to decide whether or not it should support a research project. Assume that the research project costs c units. If it succeeds, it will bring in a return of r units. If it fails, it will not bring in anything.

		Research	
		Succeeds	Fails
Company	Supports research	$r - c$	$-c$
	Neglect research	0	0

- Reformulate the payoff matrix to a **regret matrix**

		Research	
		Succeeds	Fails
Company	Supports research	0	c
	Neglect research	$r - c$	0

Minimax - One-person

- If the company supports the research and it fails, the company's regret will be c , the price of research. If the company supports the research and it succeeds, the company will have no regrets. If the company neglects research and it would have succeeded, its regret value is $r-c$, the return on the research.
- The objective is to **minimise the maximum possible regret**.

		Research	
		Succeeds	Fails
Company	Supports research	0	c
	Neglect research	$r-c$	0

Minimax - One-person

- Applying the Minimax principle:
 - If $(r-c) > c$, the company should support research.
 - If $(r-c) < c$, the company should not support research.
- In other words, the company should support research if $c < r/2$, i.e., if the expected return on research is more than twice its cost.

		Research	
		Succeeds	Fails
Company	Supports research	0	c
	Neglect research	r-c	0

Minimax - Two-person

- Example Game: A received reports that B's convoy would be heading to reinforce their troops. The convoy could take one of two routes - the Northern or the Southern route. A have to decide where to disperse their reconnaissance aircraft - in the north or the south - in order to spot the convoy as early as possible.
- The following payoff matrix shows the possible decisions made by A and B, with the outcomes expressed in the number of days of bombing A could achieve with each possibility

		B	
		North	South
A	North	2	2
	South	1	3

Minimax - Two-person

- Applying the Minimax principle: To minimise the worst possible outcome, A would have to choose the north. This ensures them 2 days of bombing, whereas they risk only 1 day of bombing if they focus on the south. Therefore, by minimax, the best strategy for A would be to focus on the north.
- B can use the same strategy. The worst possible outcome for them is the 3 days of bombing which might occur if they took the southern route. Therefore, B would take the northern route.

		B	
		North	South
A	North	2	2
	South	1	3

Minimax - Two-person

- Both A and B settled on the (North, North) square as the best outcome for both of them. Neither could do any better if the opponent was rational. In this case, the maximin and the minimax strategies produce the same result.
- **Nash equilibrium** - A set of strategies for which no players can improve their payoffs by changing their strategies.

		B	
		North	South
A	North	2	2
	South	1	3

Mixed Strategies and Randomisation

- In some cases, there is no saddle point, and the players have to choose their strategies with a degree of randomness.
- Example Game: "Matching Pennies" Two players simultaneously place a penny on a table, either heads up or tails up. If the pennies are facing the same way, player 1 gets to keep both pennies. Otherwise, player 2 gets to keep both.

		Player 2	
		Heads	Tails
Player 1	Heads	1	-1
	Tails	-1	1

Mixed Strategies and Randomisation

- There is no clearly defined strategy for each player. The best way to play is to choose the position of the coin randomly.
- If either player follows this strategy, then in the long run, the payoffs for each will be 0.
- Notice that if, say, player 1 uses a 50/50 strategy, while player 2 plays heads 75% of the time, in the long run, both players will still have payoffs of 0.
- But if player 2 follows the 75/25 strategy, then player 1 can easily take advantage of it by playing heads more frequently, and therefore winning more frequently. So, it is important for each player to not only maintain a random strategy, but to also analyse the strategy of the other player.

Maxi mini things Summary

- **Maximax** principle is ultra-optimistic, expecting the best possible payoff - choosing the best of the best possible outcomes.
- **Maximin** is ultra-pessimistic, expecting the worst possible payoff - choosing the best of the worst possible outcomes.
- **Minimax Theorem** proven by John Von Neumann
Minimax: Minimise the maximum loss
Maximin: Maximise the minimum gain
- **Nash equilibrium** is a set of strategies for which no players can improve their payoffs by changing their strategies. In general, a game may have multiple Nash equilibria.

Game Theory Terminologies

- Players of a game are said to have **perfect information** if the sequence of moves played by each player is known to all players. **Combinatorial game** is a deterministic game in which all players have perfect information e.g. chess, tic-tac-toe
- A game is said to be **solved** when the optimal strategy (if one exists) is known.
- **Zero-sum game** - sum of the payoffs for all players is constant. No player can achieve a higher payoff without decreasing the payoffs of the other players.
- **Simultaneous game** - all players move effectively simultaneously e.g. The basic form of the prisoner's dilemma
- **Sequential game** - moves are non-simultaneous e.g. Chess and tic-tac-toe

Game Theory Terminologies

- **Symmetric games** - both players have an identical choice of moves to be made simultaneously e.g. prisoner's dilemma
- **Asymmetric games** - e.g. cake cutting problem
- **Strict/weakly dominant strategy** - a strategy that is always better (or worse) than any other, regardless of other players' actions. I.e. a strategy that always yields a higher payoff than any other strategy.
- **Dynamic games** - games that have multiple, sequential in time, stages and decisions (e.g. the prisoner's dilemma is not a dynamic game as there is only one decision to make).
More from Jeremy at the end.

Game Theory Example



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<https://www.youtube.com/watch?v=kaMKlnkV7Vs>

Dynamic games - Jeremy

- **Dynamic games** are common in competitive situations in real life (e.g. generator bidding in the NZ electricity market).
- Some examples:
 - Differential games, e.g. Linear Quadratic games
 - Stochastic games (MDP)
- The Nash equilibrium is the set of strategies of all players for which no single player can benefit by deviating from. This may or may not lead to a system equilibrium.
- In industry, Stackelberg games are often used (leader-follower). These are considered **dynamic** in the game theory sense but **not** in the optimization-based control sense (no states).

Exercises

- Look up the terms that are new to you in this lecture and understand them.
- Are there anything that you can think of that can be formulated as a game? What are the outcomes of using the different maxi, mini strategies?

Thanks!